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Пр. 3.

① Найти частные производные.

$$a) u = x^y + (xy)^z + z^{xy}$$

$$u'_x = y x^{y-1} + z x^{z-1} y^{z-1} + z^{xy} \ln z^y$$

$$u'_y = x^y \ln x + z y^{z-1} + z^{xy} \ln z^x$$

$$u'_z = (xy)^z \ln(xy) + xy \cdot z^{xy-1}$$

$$2) u = \frac{y}{z} + \operatorname{arctg} \frac{z}{x} + \operatorname{arctg} \frac{x}{z} \quad (\operatorname{arctg} x)' = \frac{1}{1+x^2}$$

$$u'_x = \frac{1}{(1 + \frac{z^2}{x^2}) x^2} + \frac{1}{(1 + \frac{x^2}{z^2}) z} = 0$$

$$u'_y = \frac{1}{z}$$

$$u'_z = y \cdot (-1) \cdot \frac{1}{z^2} + \frac{1}{(1 + \frac{z^2}{x^2}) x} + \frac{1}{(1 + \frac{x^2}{z^2}) z^2} = 0$$

② Тест. произв. II порядка:

$$a) f(x, y) = x^4 - 2x^2 y^3 + y^5 + 1$$

$$f''_{xx} = (4x^3 - 2y^3 \cdot 2x)' = 12x^2 - 4y^3$$

$$f''_{yy} = (-2x^2 \cdot 3y^2 + 5y^4)' = 20y^3 - 12x^2 y$$

$$f''_{xy} = (4x^3 - 4x \cdot y^3)'_{\text{по } y} = -12xy^2 = f''_{yx}$$

$$b) f(x, y) = \frac{\cos y^2}{x}$$

$$f'_x = \cos y^2 \cdot (-1) \cdot \frac{1}{x^2}$$

$$f'_{xx} = +\cos y^2 \cdot (+2) \cdot \frac{1}{x^3}$$

$$f'_y = \frac{-1}{x} \cdot ((\sin y^2) 2y)'$$

$$f'_{yy} = \frac{-1}{x} (\cos y^2 \cdot 2y \cdot 2y + \sin y^2 \cdot y) = \\ = -\frac{1}{x} (\cos y^2 \cdot 4y^2 + \sin y^2 \cdot y)$$

$$f''_{xy} = f''_{yx} = (-\cos y^2 \cdot \frac{1}{x^2})' = \frac{\sin y^2 \cdot 2y}{x^2}$$

③ Решить $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$

$$f(x, y) = xy \sqrt{9 - x^2 - y^2} = xy (9 - x^2 - y^2)^{\frac{1}{2}}$$

$$f'_x = y (1 \sqrt{9 - x^2 - y^2} + x \cdot \frac{1}{2} (9 - x^2 - y^2)^{-\frac{1}{2}} \cdot (-2x)) =$$

$$= y \left(\frac{9 - x^2 - y^2 - x^2}{\sqrt{9 - x^2 - y^2}} \right) = y \left(\frac{9 - 2x^2 - y^2}{\sqrt{9 - x^2 - y^2}} \right) = 0 \quad \begin{cases} y = 0 \\ 9 - 2x^2 - y^2 = 0 \\ \Downarrow \\ y^2 = 9 - 2x^2 \end{cases}$$

$$\frac{\partial f}{\partial y} = x \sqrt{9 - x^2 - y^2} + xy \cdot \frac{1}{2 \sqrt{9 - x^2 - y^2}} (-2y) = x \left(\frac{9 - x^2 - 2y^2}{\sqrt{9 - x^2 - y^2}} \right) = 0$$

$$\begin{cases} x = 0 \\ x^2 = 9 - y^2 \end{cases}$$

$$\Delta f = f(x + \Delta x) - f(x)$$

$$\Delta_x f(x, y) = f(x + \Delta x, y) - f(x, y) \quad - \text{частное приращение по } x.$$

$$\Delta_y f(x, y) = f(x, y + \Delta y) - f(x, y) \quad - \text{по } y$$

$$\Delta f = f(x + \Delta x, y + \Delta y) - f(x, y)$$

④ Частное и полное приращ. в т. (3; -2).

$$\Delta x = 0,1 \quad \Delta y = -0,05$$

$$f(x, y) = xy^2 - \frac{x}{y}$$

$$\Delta_x f(x, y) = f(x + 0,1, y) - f(x, y) = (x + 0,1)y^2 - \frac{x + 0,1}{y} = 0,1y^2 - \frac{0,1}{y} = 0,1 - 0,05 = 0,05$$

$$\Delta_y f(x, y) = f(x, y - 0,05) - f(x, y) = x(y - 0,05)^2 - \frac{x}{y - 0,05} - xy^2 + \frac{x}{y} =$$

$$= x \cancel{y^2} - x(y - 0,05)^2 - \frac{x}{y - 0,05} + \cancel{xy^2} + \frac{x}{y} = x \cdot 3 \cdot 2 + 3 \cdot 0,75 + \frac{30}{2 + 0,5} + \frac{-1,5}{-2} = 6,45$$

⑤ Частн. произв., частные и

$$a) z = \sqrt[3]{x+y^3} = (x+y^3)^{\frac{1}{3}}$$

$$f'_x = \frac{1}{3\sqrt[3]{(x+y^3)^2}}$$

$$f'_y = \frac{3y^2}{3\sqrt[3]{(x+y^3)^2}}$$

$$b) z = \cos \frac{x^2+y^2}{x^3+y^3}$$

$$f'_x = -\sin \frac{x^2+y^2}{x^3+y^3} \cdot \left(\frac{2x \cdot (x^3+y^3) - (x^2+y^2) 3x^2}{(x^3+y^3)^2} \right)$$

⑥ a) (1,1)

$$f(x,y) = x^2 y$$

$$\cancel{f}(1+\Delta x, 1+\Delta y) - f(1,1) = (1+\Delta x)^2 (1+\Delta y) - 1 = (1+2\Delta x+\Delta x^2)(1+\Delta y) - 1 =$$

$$= \cancel{1} + \Delta y + 2\Delta x + 2\Delta x \Delta y + \Delta x^2 + \Delta x^2 \Delta y - \cancel{1}$$

$$f'_x = 2xy$$

$$f' = 2xy + 2x^2$$

$$f'_y = x^2$$

$$f'(1+\Delta x, 1+\Delta y) - f(1,1) = 2(1+\Delta x)(1+\Delta y) + (1+\Delta x)^2$$